

Chapter 21

Nuclear chemistry

1. Which one of the following is a correct representation of a beta^{e-} particle? D.

A) ${}^4_2\text{e}^-$

B) ${}^1_0\beta^-$

C) ${}^0_1\text{e}^-$

D) ${}^0_{-1}\text{e}^-$

E) ${}^2_4\beta^-$

$$\begin{matrix} 0 \\ 1 \end{matrix}$$

2. The mode of decay of $^{32}_{15}\text{P}$ is $^0_{-1}\beta$ B $^{32}_{16}\text{S}$.

A) alpha emission

☒ B) beta emission

C) positron emission

D) electron capture

E) neutron capture

3. What is emitted in the nuclear transmutation, ${}_{13}^{27}\text{Al} (n,?) {}_{11}^{24}\text{Na}$?
 A .

$$\begin{array}{c} 27 \\ 13 \end{array} + \begin{array}{c} 1 \\ 0 \end{array} = \begin{array}{c} 24 \\ 11 \end{array} + \begin{array}{c} 4 \\ 2 \end{array}$$

- A) an alpha particle
- B) a beta particle
- C) a neutron
- D) a proton
- E) a gamma photon

4. The beta decay of cesium-137 has a half-life of 30.0 years. How many years must pass to reduce a 25 mg sample of cesium 137 to 8.7 mg? A

- A) 46
- B) 32
- C) 3.2
- D) 50
- E) 52

5. Which one of the following devices converts radioactive emissions to light for detection? C

A) Geiger counter

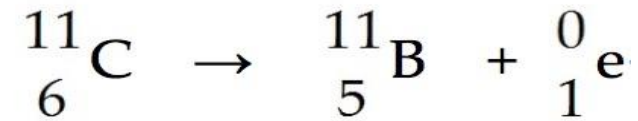
B) photographic film

C) scintillation counter

D) radiotracer

E) none of the above

6. Carbon-11 decays by positron emission:



The decay occurs with a release of 2.87×10^{11} J per mole of carbon-11.

When 5.00 g of carbon-11 undergoes this radioactive decay, A g of mass is converted to energy.

A) 1.45×10^{-3}

B) 4.35×10^5

C) 1.45×10^{-6}

D) 6.90×10^2

E) 1.59×10^{-2}

7. The mass of a proton is 1.00728 amu and that of a neutron is 1.00867 amu. What is the binding energy per nucleon (in J) of a ^{59}Co nucleus? A (The mass of a cobalt-59 nucleus is 58.9332 amu.)

- A) $1.37 \times 10^{-12} \text{ J}$
- B) $4.87 \times 10^{13} \text{ J}$
- C) $1.34 \times 10^{-34} \text{ J}$
- D) $2.50 \times 10^{12} \text{ J}$
- E) $4.80 \times 10^{10} \text{ J}$

$$\Delta m = 0.5408 \text{ amu} = 0.5408 \times 1.66 \times 10^{-27} \text{ kg}$$

$$\Delta E = mc^2 = 8.98 \times 10^{-28} \times 9 \times 10^{16} = 8.08 \times 10^{-11}$$

$$\Delta E / \text{nucleon} = \frac{8.08 \times 10^{-11}}{59} = 1.37 \times 10^{-12}$$